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1.7.ER Diagram & Schema Design

The Entity-Relationship (ER) Diagram serves as the visual blueprint for the entire system, illustrating how different data points interact. In our design, the Flight entity has a "one-to-many" relationship with the Seat entity, as one flight contains many individual seats. Similarly, the Passenger entity has a "one-to-many" relationship with Bookings, allowing a single traveler to manage multiple upcoming trips. The schema design follows the principles of Third Normal Form (3NF) to eliminate "update anomalies" and "redundant data." For instance, instead of storing the airline's phone number in every flight record, we store it once in the Airlines table and reference it via an Airline_ID. This normalization process ensures that if an airline changes its contact info, we only need to update it in one location. We also use a "junction table" for scenarios where many-to-many relationships exist, such as passengers and their various travel documents. The schema includes specific data types for every attribute, such as DATETIME for departures and DECIMAL for currency, to maintain precision. By strictly adhering to these design patterns, the database remains lean and efficient. This structured approach prevents the "data bloat" that often slows down older, poorly designed airline systems. The resulting schema is both easy for developers to understand and optimized for the database engine to execute

